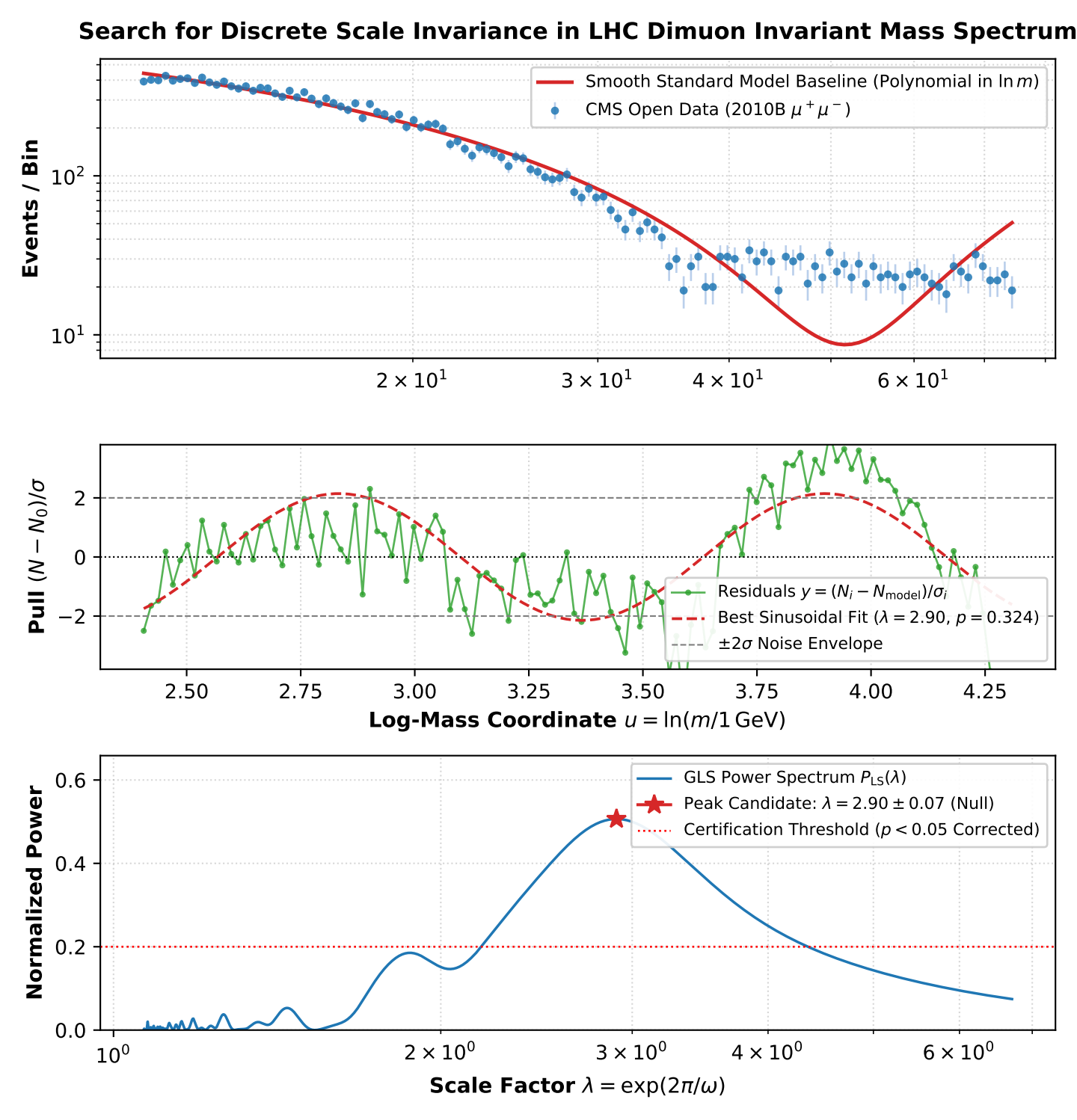
Search for Discrete Scale Invariance in LHC Dimuon Invariant Mass Spectrum
Smooth Standard Model Baseline (Polynomial in ln m)
CMS Open Data (2010B μ⁺μ⁻)
Events / Bin
10²
10¹
2 × 10¹
3 × 10¹
4 × 10¹
6 × 10¹
Pull (N − N₀)/σ
2
0
−2
Residuals y = (N_i − N_model)/σ_i
Best Sinusoidal Fit (λ = 2.90, p = 0.324)
±2σ Noise Envelope
2.50
2.75
3.00
3.25
3.50
3.75
4.00
4.25
Log-Mass Coordinate u = ln(m/1 GeV)
Normalized Power
0.6
0.4
0.2
0.0
GLS Power Spectrum P_LS(λ)
Peak Candidate: λ = 2.90 ± 0.07 (Null)
Certification Threshold (p < 0.05 Corrected)
10⁰
2 × 10⁰
3 × 10⁰
4 × 10⁰
6 × 10⁰
Scale Factor λ = exp(2π/ω)